

27	A photon of light has frequency f, momentum p, and speed c. The Planck constant is h. Which expressions for the momentum and the energy of the photon are correct?			
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		momentum	energy	
	A	hf	pc	
	B	hf	$\frac{pc}{2}$	
	C	$\frac{hf}{c}$	pc	
	D	$\frac{hf}{c}$	$\frac{pc}{2}$	

Answer: C

Explanation/Working:

de Broglie equation: $\lambda = \frac{h}{p}$

$$\frac{c}{f} = \frac{h}{p}$$

$$p = \frac{hf}{c}$$

Since $E = hf$

$$p = \frac{E}{c}$$

$$E = pc$$

28	An electron in an atom transitions from -25.0 eV to -80.0 eV . A photon is emitted in the process.			
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